

Module 3 — The Query Library

45-minute block · TIP4PATLIBS course material

How to read this document. The running text addresses **you, the participant**. Boxes labelled **Trainer** are for whoever is **running the session**; boxes labelled **Trap** are traps that have caught people before.

Learning objective

I can pick a ready-made PATSTAT query, adapt its parameters to my own question, and judge whether the result is plausible — without writing SQL.

Prerequisites

- **Module 1** — TIP is running, PATSTAT connects with `PatstatClient(env='PROD')`.
- No SQL knowledge. That is the point of this module.

Sub-objectives

By the end you can:

1. **Map a client's question onto a query category** — decide *which* of the library's queries answers what you were actually asked.
2. **Change parameters safely** — country, year window, technology field — and know which changes are cheap and which are expensive.
3. **Judge plausibility** — recognise a result that is suspiciously large, suspiciously small, or silently truncated.
4. **Know when to stop** — recognise the questions the library cannot answer, and what to reach for instead.

Material

Folder	2_querylib/
Notebooks	TIP_for_PATLIBs_QueryLib.ipynb (the browser) · TIP_for_PATLIBs_InteractiveQueryDemo.ipynb (point-and-click)
Runs on	EPO TIP, PATSTAT PROD

Phase 1 · Introduction (≈ 7 min)

The question this module answers

A client walks in: "Who in Europe is working on solid-state batteries?"

Give the room two minutes to say how they would answer it. Collect the proposals.

What almost always comes back first is a keyword search. And a keyword search is not wrong — it is just not defensible. Ask the follow-up: *how would you prove to that client that your list is complete?* Keywords miss everything phrased differently, and catch everything that merely mentions the term. Neither problem is visible in the result.

That gap is what a curated query library exists to close. Someone has already decided which tables to join, which filters are non-negotiable, and what "complete" means for each type of question. Your job shifts from *writing* the query to *choosing* it and *reading* it.

Teaching and learning activity		Time
Opening	Trainer poses the client question; room proposes approaches out loud	2 min
Tension	Trainer asks "how do you prove it is complete?" — the proposals fall apart	2 min
Framing	Trainer introduces the library as <i>pre-made professional decisions</i> , not as a shortcut	3 min

Trainer. The expected wrong answer is *"search Google Patents"* or *"type keywords into Espacenet"*. Take it seriously — it is what most people do and it is often adequate. Do not mock it; ask what happens when the client's competitor turns up a patent that your list missed. The room will get there on its own.

Trainer. If the room is quiet, name a concrete failure: a search for *"battery"* misses every document that says *"accumulator"* or *"electrochemical cell"*.

Phase 2 · Working through (≈ 28 min)

Step 1 — Open the library and look at what is there (8 min)

Run `TIP_for_PATLIBs_QueryLib.ipynb` top to bottom. It has only two cells: a setup cell that initialises the registry, and the **Query Browser**.

In the browser:

- **Browse by category** using the dropdown.
- **Search** by typing a keyword — it searches titles, descriptions and tags.
- **Preview** a query to see what it needs and what it returns.

- **View SQL** to see the template with its parameters highlighted.

Do this deliberately: open "View SQL" at least once. You are not expected to read it fluently. You are expected to see that there *is* something to read — that the number the library gives you comes from a statement you could inspect, hand to a colleague, or argue about.

Trainer. This is the moment worth slowing down for. The difference between a tool and a black box is whether you can open it. Say it out loud: *an analysis you cannot open up is one you cannot defend*. The same sentence returns in modules 4 and 8.

Step 2 — Adapt one query to your own question (12 min)

Pick a query that matches a question you actually get asked. Change its parameters — country, year window, technology field — and run it.

Two rules while you do this:

- **Change one thing at a time.** If the result surprises you, you want to know which change caused it.
- **Watch the clock on the query.** The library reports how long each query took.

Trap — PATSTAT PROD is not fast. A query scanning a full year of filings takes tens of seconds to a minute or more. That is normal, not a fault. Widening a year window or dropping a country filter multiplies the work; a query that ran in 10 seconds can take several minutes with the filter removed. Say this to the room *before* someone decides the notebook has crashed.

Step 3 — The interactive demo (8 min)

Open `TIP_for_PATLIBs_InteractiveQueryDemo.ipynb` and run the first cell to load the components. Then use the **Selection** controls — jurisdiction, region, technology field or custom IPC/CPC, date range — and read the **Results**: a table and a chart, no code touched.

This is the same data through a different door. Note which door you would put in front of a client, and which you would use yourself.

Trap — Stop after the "Data Validation" section. The notebook ends with two cells headed "Story 3.2 AC Validation" and "Story 3.3 AC Validation". Those are developer acceptance tests left over from building the tool — not course content. Skip them; they will run, and their output means nothing to a participant.

Trainer. Note this as a known rough edge rather than apologising for it. It is honest, and it is on the cleanup list.

Step	What you do	What you see	Time
1	Browse the library, open one query's SQL	The decisions someone made for you	8 min
2	Adapt parameters, run, time it	Your own question answered — and what it costs	12 min
3	Drive the interactive demo	The same data through a point-and-click door	8 min

Phase 3 · Learning outcome (≈ 10 min)

What now exists

- One query adapted to a question **you** are actually asked, with its result table.
- A first feel for what a PATSTAT query costs in time.

Self-check

Answer these before moving on. If one is unclear, go back to the query that raised it.

1. **Which query would you use** for *"has patenting in my region gone up or down over ten years?"* — and which would you use for *"who are the biggest filers in this technology?"*
2. **Why does the same question return fewer rows** when you filter `appln_auth = 'EP'` than when you count `docdb_family_id`? (*Because one counts filings at one office, the other counts inventions wherever they were filed. Module 4 makes this the centre of the lesson.*)
3. **You widened the year window from 5 to 20 years and the query is still running after three minutes.** Broken, or working? (*Working. Narrow it, get the answer, then widen deliberately.*)
4. **A client asks which of two companies has the "better" portfolio.** Can the library answer that? (*No. It can measure size, reach and technology mix. "Better" is a judgement — and module 8 is about exactly that boundary.*)

Transfer to your own work

Write down the **three questions you are asked most often** at your PATLIB desk. For each, note which library query comes closest, and what is missing. That list is your agenda for modules 4 and 5 — and it is worth keeping.

Trainer. Close by collecting two or three of those questions from the room. If one of them is *"who around here is patenting?"*, you have your bridge into module 5. If it is *"is this company big or small in IP?"*, that is module 4.

Where this leads

Next	Why
Module 4 — PATSTAT Explorer	The library counts <i>names</i> . Module 4 shows why one company is forty different names, and how to consolidate them.
Module 5 — Lead Generation	The first full application: a region's applicants, ranked and segmented.

Notes for the next revision

- The two "*Story 3.x AC Validation*" cells at the end of the interactive demo are developer scaffolding and should be removed from the course version of the notebook.
- Module 3 is the shortest module in the course (≈ 8 minutes of notebook content). This block works because the introduction and the transfer exercise carry real weight — do not cut them to save time.